AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 8 · STUDENT LESSON

Leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the claim. Group the source. Protect the final check.

Audit related-source and prediction-time leakage without treating a repaired split as a model result.

Keep one habit: Complete A–F plan → group audit → plant/view matrix → four leakage paths → independent five-plant repair.

MAIN LESSON · SECTION 1

Keep evaluation evidence outside the learning decisions

> Different filenames do not make related observations independent.

Learning goals

I can choose a split unit that matches the intended **prediction** claim.

I can keep related examples together and audit **group** overlap.

I can record split roles counts and limits before model evaluation.

Recall label quality and evaluation roles

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.
- Why does agreement between two annotators not prove their labels are correct?

▼ **Optional readiness · Stored training neighbours**

In the optional extension, nearest means the smallest stated distance to a stored **training** example. Why must validation examples stay outside that stored **training** pool?

> **My readiness explanation**

End goal

By the end, I can explain **leakage** using a traced example and a stated limitation.

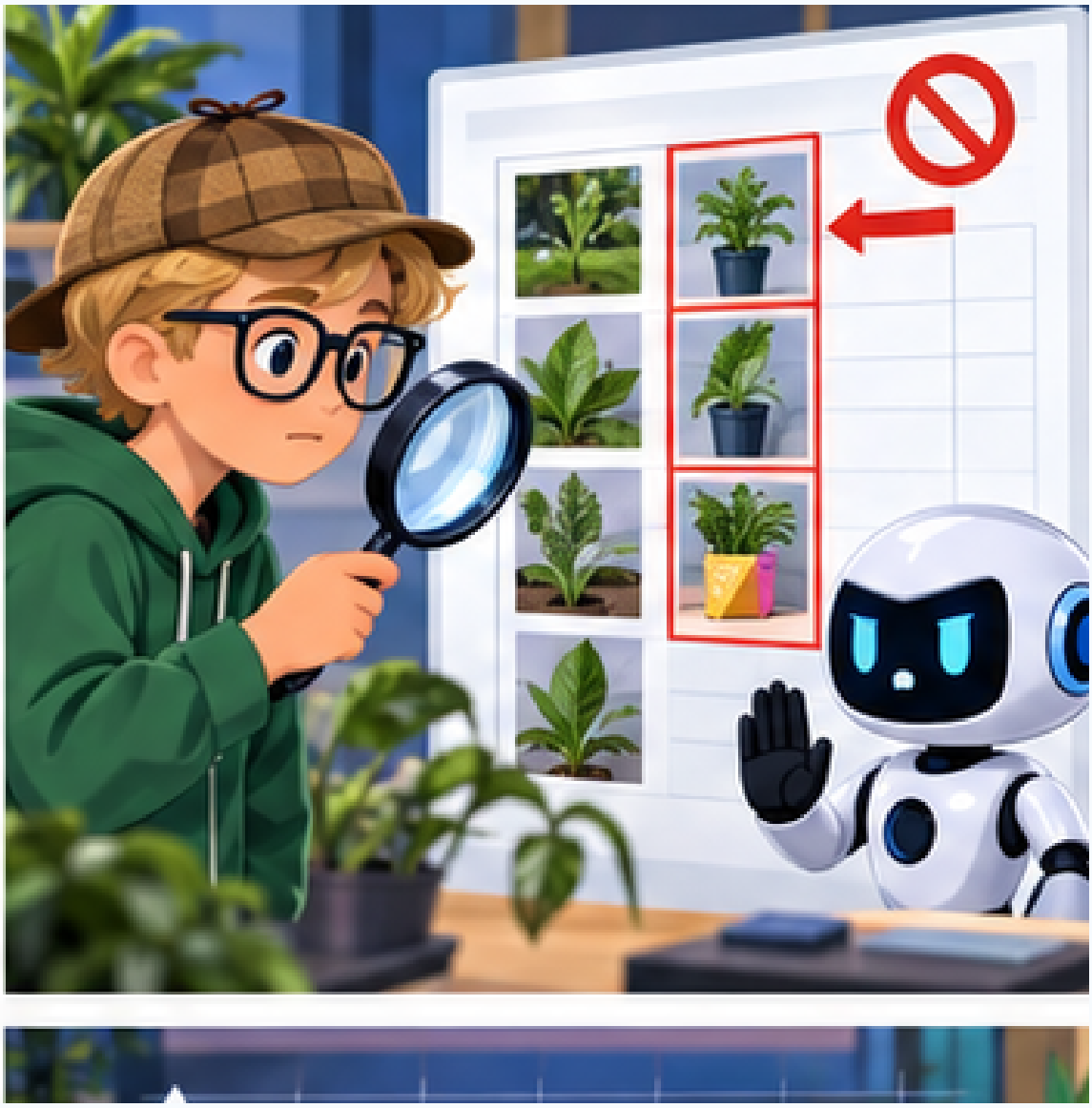
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Leakage occurs when a feature exposes the target through information that would be unavailable, created later, or derived from the answer at real **prediction time**.



LEARNING TARGET Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Scaling without leakage

This week's end goal

MAIN LESSON · SECTION 2

Choose groups that match the evaluation question

> For new trips on the same route, hold out whole trips.

Scenario and intended claim

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip.

They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the **group** is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions.

We therefore keep all frames from a trip in one set. To claim success on new routes, **group**ing only by trip would not be enough.

All twelve frame assignments

Trip group	Frame IDs	Assigned role
A	A1 A2 A3	Training

Trip group	Frame IDs	Assigned role
B	B1 B2	Training
C	C1	Training
D	D1 D2	Validation
E	E1 E2	Test
F	F1 F2	Test

Source and metadata limitations

This small fictional assignment is for learning split design, not **evidence** of real performance. Each frame belongs to one **group** and one set.

Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate **prediction input**.

▼ Optional extension · Training-only scaling and fixed nearest neighbour

This optional extension introduces a fixed one-nearest-neighbour example; the main nearest-neighbour lesson comes later in Week 12. With two features, their numerical **range**s can give one feature much more influence. Scaling changes those contributions. We will learn a minimum and **range** for each feature from **training** only, then reuse them for every example.

These fictional trips have a distance reading in cm, a brightness reading in sensor units and a reviewed lane label. Features are available before **prediction**; labels are separate **evidence**.

Related trips have already been **group**ed before splitting. For this exercise use k=1, add absolute differences across the two features, and break equal distances by alphabetical ID.

ID and label	Distance cm	Brightness units
A clear	10	200
B blocked	20	100

ID and label	Distance cm	Brightness units
C blocked	20	200

For query (11,120), raw distances to A,B,C are 1+80=81, 9+20=29 and 9+80=89. B wins and predicts blocked. Adding these different units is an unscaled numerical choice; the large brightness **range** strongly affects it.

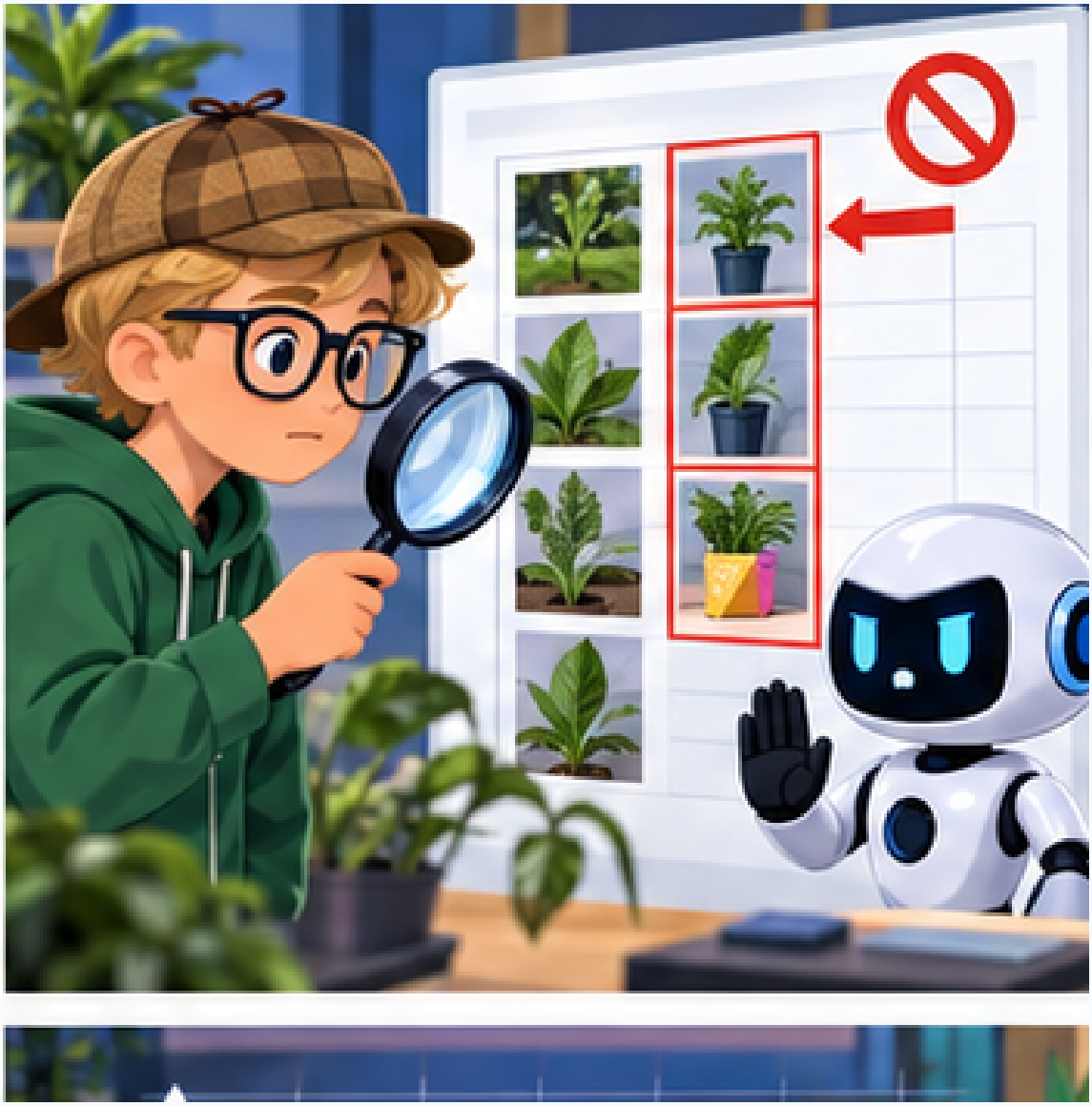
Scaling offers another explicit distance choice, not proof of which features matter.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **leakage** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Scaling without leakage

MAIN LESSON · SECTION 3

Audit the group sets before fitting anything

> Disjoint listed groups are necessary here, but do not prove the source assignments are complete.

Training, validation and final test roles

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the **test set** for a final check after those choices are fixed.

A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

Complete group-overlap script

```
train_groups = {"A", "B", "C"}
valid_groups = {"D"}
test_groups = {"E", "F"}
shared = ((train_groups & valid_groups)
          | (train_groups & test_groups)
          | (valid_groups & test_groups))
print(sorted(shared))
print(len(train_groups), len(valid_groups),
      len(test_groups))
```

Trace group counts and assumptions

The first line of **output** is []: no trip ID is shared between any pair of sets. The second is 3 1 2: **training**, validation and test contain three, one and two trip **group**s. The symbol & finds shared values and | combines sets.

These are **group** counts, not frame counts.

Keep each trip in one role

WHOLE GROUPS	ROLE	PURPOSE
A B C with all frames	Training	Fit from examples
D with all frames	Validation	Compare choices
E F with all frames	Test	Final check after choices

Keep related frames together. Separate group IDs prevent this form of overlap; they do not prove that every other source of leakage is absent.

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer.

Check the assignment table and source records too.

▼ **Optional extension · Fit and reuse one training scale**

```
train = [(10, 200), (20, 100), (20, 200)]
low = [min(row[j] for row in train) for j in range(2)]
span = [max(row[j] for row in train) - low[j]
        for j in range(2)]
if 0 in span:
    raise ValueError("Review a constant training feature")

def transform(row):
    return tuple((row[j] - low[j]) / span[j]
                 for j in range(2))

scaled_train = [transform(row) for row in train]
query = (11, 120)
print(low, span)
```

```
print(scaled_train)
print(transform(query))
```

For each feature use scaled value = (value minus **training** minimum) divided by **training range**. **range**(2) visits feature positions 0 and 1.

The code prints [10, 100] [10, 100], then [(0.0, 1.0), (1.0, 0.0), (1.0, 1.0)], then (0.1, 0.2). Each minimum becomes 0 and each maximum becomes 1.

The first calculations fit low and span from train. **transform** uses those stored values without learning new ones. The code assumes nonempty **training** rows with exactly two finite numeric features; it does not check all **input**s.

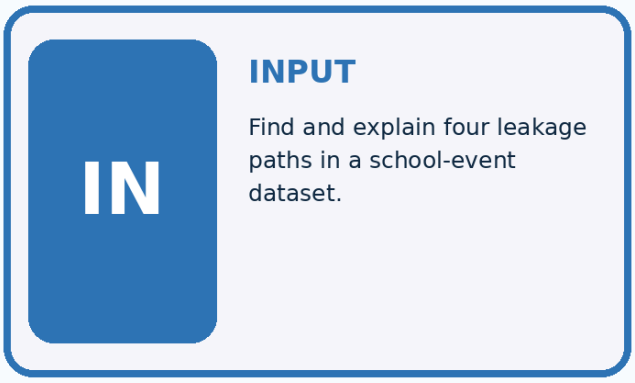
Its explicit zero-**range** guard stops division by zero so a constant feature can be reviewed. This is a classroom policy, not a claim that every library stops this way.

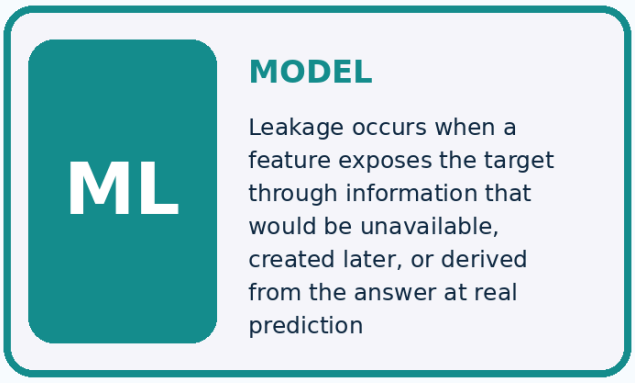
▼ Earlier wording, alternatives and picture transcripts

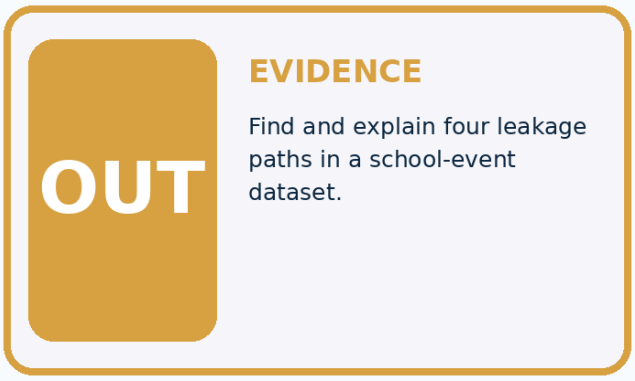
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Find and explain four **leakage** paths in a school-event dataset.

MODEL IDEA **Leakage** occurs when a feature exposes the target through information that would be unavailable, created later, or derived from the answer at real **prediction time**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD A leaf file name contains 'diseased'—the answer is hidden in the **input**. Treatment outcome was recorded after diagnosis, so it is unavailable at **prediction time**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Scaling without leakage

Connect the picture and the explanation

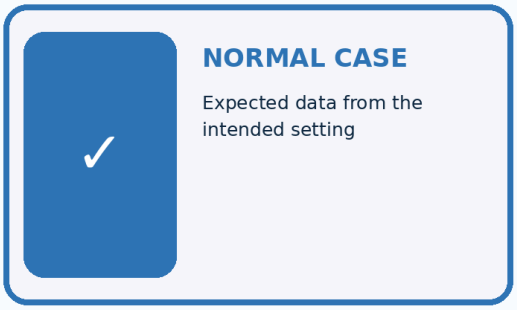
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

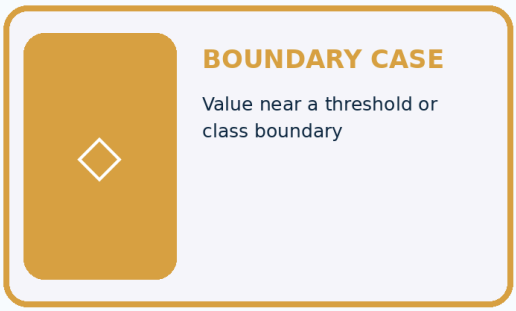
Compare related-source overlap with a valid new-source holdout

> Audit the source relation rather than trusting a random seed or unique filenames.

Four evaluation conditions



NORMAL Expected data and expected behavior.

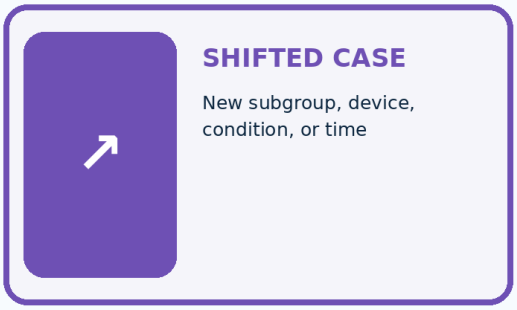


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, sub**group**, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Move Fir’s first view into **training** while retaining its other two in test.

▼ **Check after predicting**

Prediction: The unseen-plant evaluation is compromised.

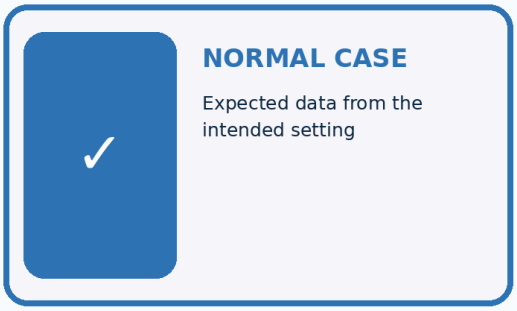
Reason: Fir’s identity now appears in **training**; different view IDs do not remove the shared plant relationship.

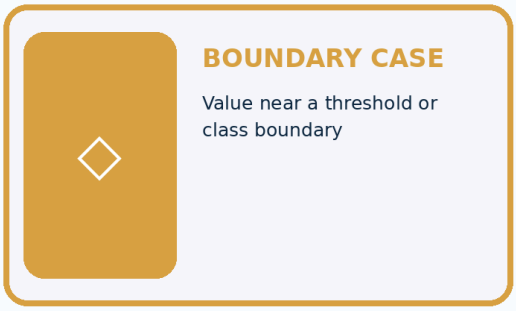
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

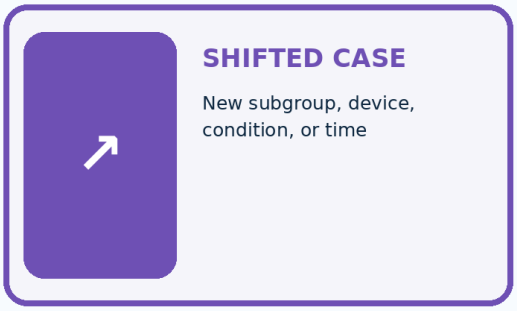
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, sub**group**, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read source groups through both the picture and assignment grid

> A row count is different from a count of independently held-out source identities.

Complete plant facts and scope

Split by plant identity when testing generalisation to new plants.

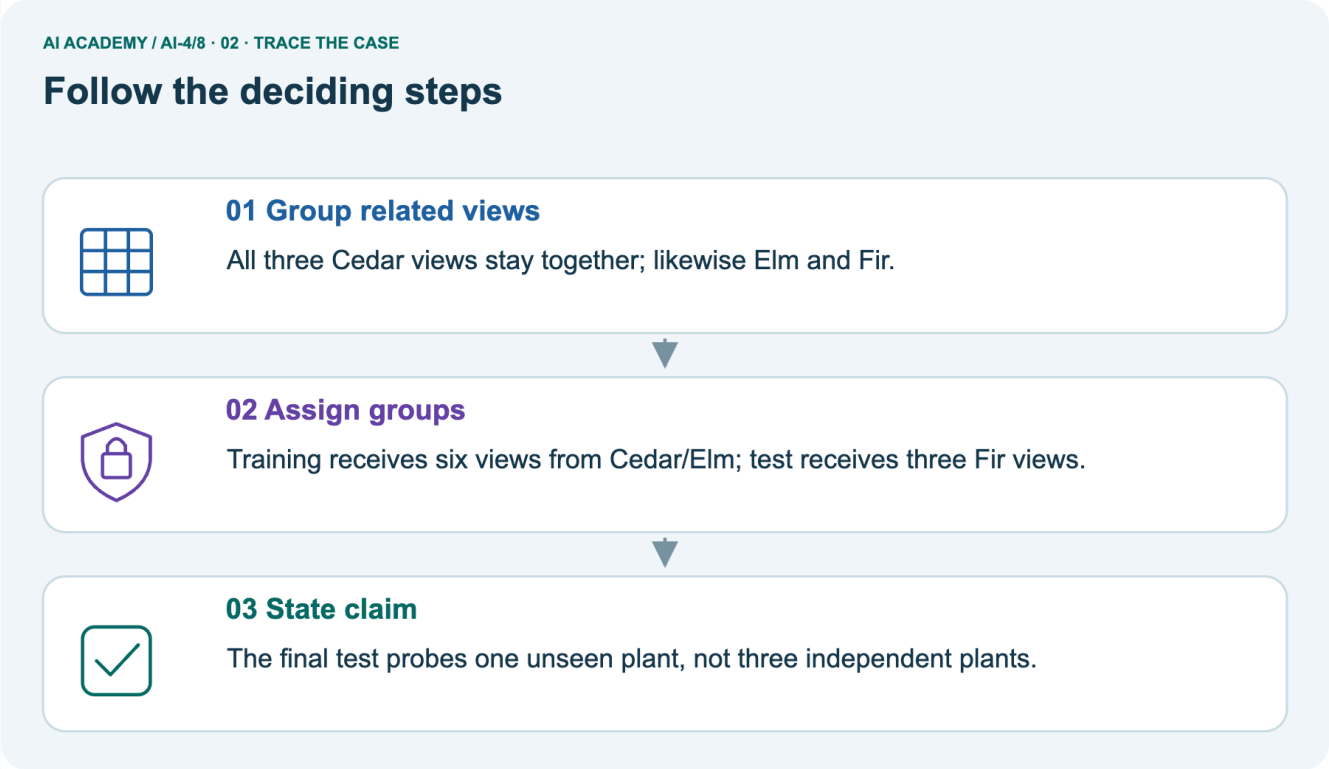
Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The trip plan and plant picture have different counts. Trips A–F: training6 frames/3 trips, validation2/1, test4/2. Moving D2 to training gives7/1/4 frames and4/1/2 listed trip memberships, with D shared by training and validation. Plant picture: Cedar/Elm train6views/2plants, Fir test3views/1plant.

Moving Fir view1 to training gives7train/2test views and shared Fir; there is no completely unseen plant identity in that remaining test. No score change is measured by either illustration.

Trace source identities and view counts



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

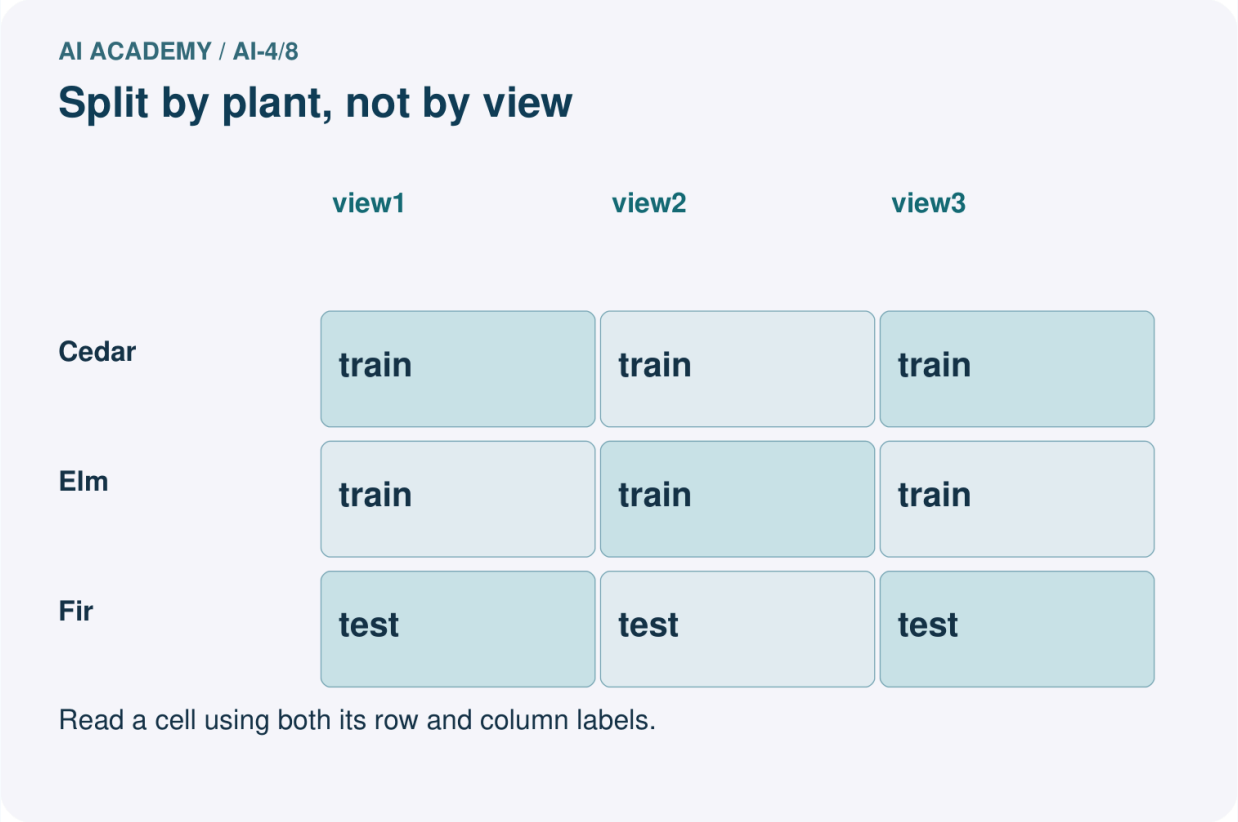
STEP	WHAT HAPPENS AND WHY
1. Group related views	All three Cedar views stay together; likewise Elm and Fir.
2. Assign groups	Training receives six views from Cedar/Elm; test receives three Fir views.
3. State claim	The final test probes one unseen plant, not three independent plants.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original plant-by-view matrix

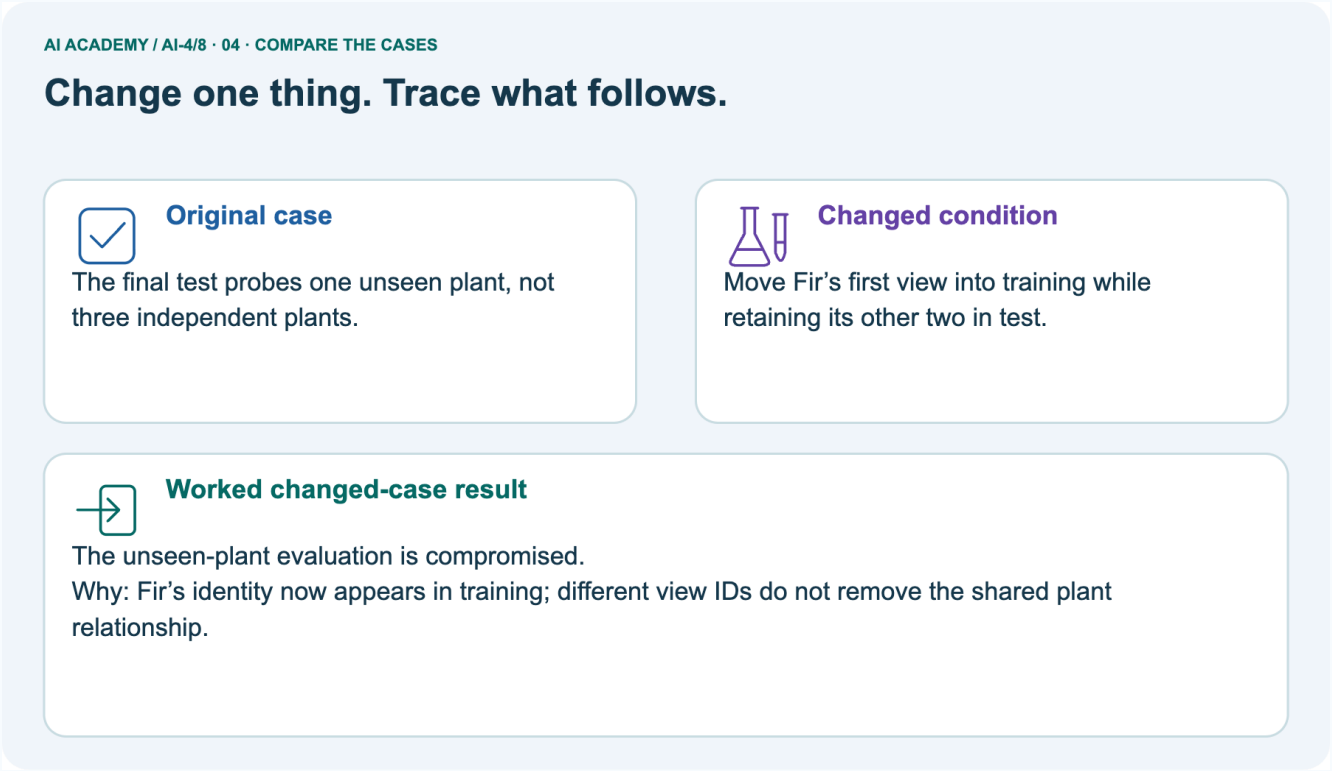


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict Fir crossing roles

- Change one thing: Move Fir's first view into **training** while retaining its other two in test. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the tiny-test limit

- A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

- Read the labelled picture: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the full trip plan and D2 change

Suppose A1 and A2 are in **training** while A3 is in test. The frame IDs differ, but A appears in both sets. Shuffling the rows does not remove this relationship.

For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Audit that every source frame occurs exactly once, no **group** crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six **training**, two validation and four test: twelve total.

A fixed random seed can reproduce a random split, but it cannot make a flawed **group**ing choice fair.

> **A learner moves D2 from validation to training but leaves D1 in validation. Give the new row counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.**

A split record names the **prediction** goal, source version, **group** definition, exact assignments, row and **group** counts, date and any random method or seed used. For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 6.

▼ Optional extension · Reuse a fixed scale

One stored scale for every split

EXAMPLES	OPERATION	STORED VALUES
Training only	Fit minima and ranges	low 10 100 span 10 100
Training examples	Transform each row	Reuse low and span
Validation and new	Transform each row	Reuse low and span

Only training examples determine the scale. Transform validation and new examples using the same stored feature order, minima and ranges.

The scaled query is (0.1,0.2). Distances to scaled A,B,C are 0.1+0.8=0.9, 0.9+0.2=1.1 and 0.9+0.8=1.7. A now wins and predicts clear. We changed feature contributions; we did not change any reference label.

Compare scaled queries with scaled **training** rows, never with raw **training** rows.

> **Show why raw distance selects B but scaled distance selects A for (11,120). Which stored numbers must remain fixed when the next query arrives?**

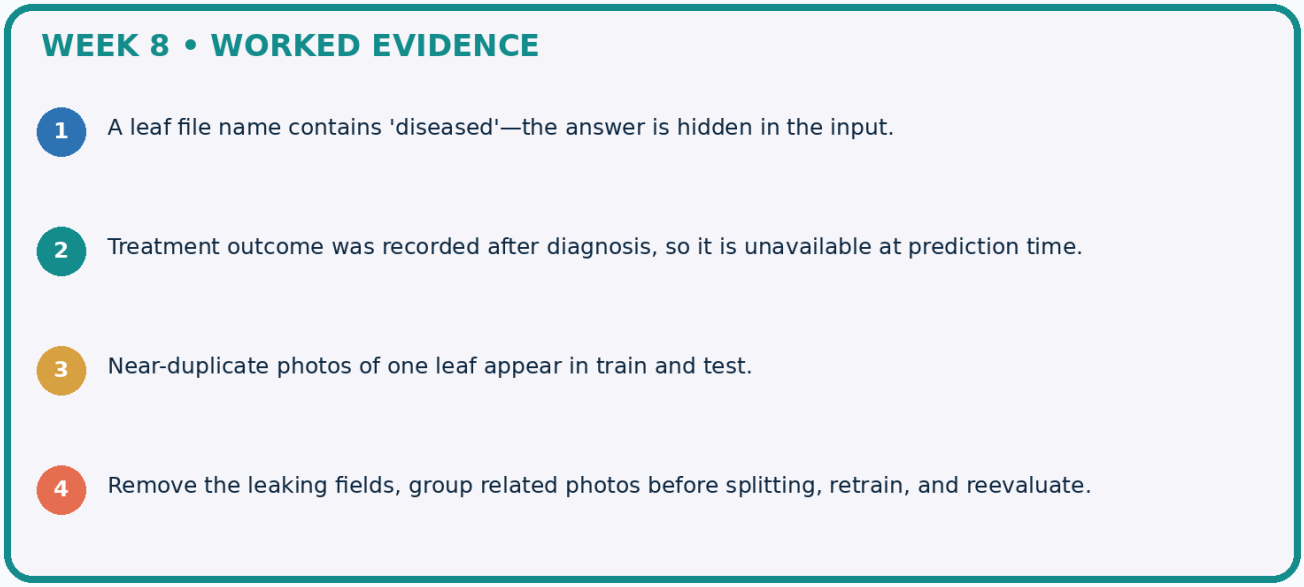
Equal **training range**s after scaling do not establish equal real-world importance. Outliers affect minima and **range**s. Scaling does not repair wrong labels, missing information or an unsuitable distance measure.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 A leaf file name contains 'diseased'—the answer is hidden in the **input**.

STEP 2 Treatment outcome was recorded after diagnosis, so it is unavailable at **prediction time**.

STEP 3 Near-duplicate photos of one leaf appear in train and test.

STEP 4 Remove the leaking fields, **group** related photos before splitting, retrain, and reevaluate.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

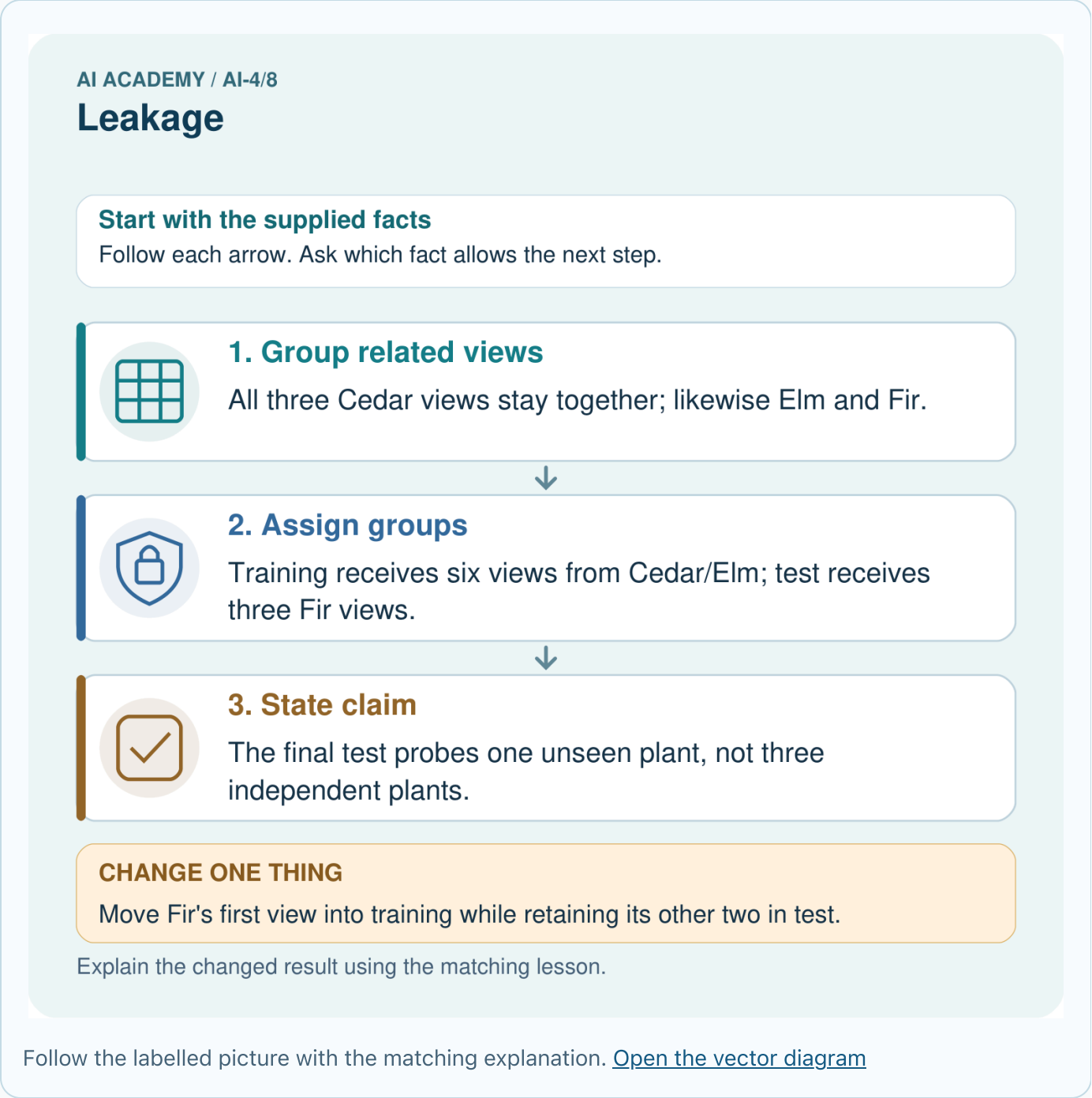
Apply the connected idea: Scaling without leakage

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Leakage

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find information that crosses the intended boundary

> Group overlap and preprocessing leakage need different repairs.

Main diagnosis and limitation

- Fit learned pre**process**ing, such as scaling values using an average, on **training** data and apply the stored **transform**ation to the other sets.

  Using the full dataset to learn such values would let held out information influence development. We will implement pre**process**ing later; **group** separation alone does not solve that issue.

- Record exclusions and label-review policies before scoring. Do not remove difficult test cases because they reduce the score.

  If a genuine data error requires repair, document it and its effect on the evaluation rather than claiming the old test stayed untouched. Next week you will build a simple baseline using these split roles.

> **All trip IDs are separate, but every trip uses the same route. Can you claim the model works on unseen routes? Explain the limitation and propose a split unit for that new claim.**

Challenge the leakage shortcut

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A highly accurate feature should be kept even when it would not exist during real use.

▼ Optional scaling contract and zero-range policy

Record the split, feature names and units, feature order, minima, **range**s, constant-feature policy, distance, k and tie rule. Keep the scale with the stored **training** examples. New data uses the same calculation. If development choices change, document them and reserve a separate final test. Later classifier-pipeline lessons combine these steps; next week records dataset contents and limitations.

If a feature is constant in **training**, its **range** is zero. This example stops for review.

One documented option is to omit that feature everywhere and update the recorded feature list; do not divide by zero or invent a scale using validation values.

Library policies differ, so inspect and record the one actually used.

> **A learner finds minima and ranges using all rows, then splits them and says there is no leakage because labels were never used. Explain the error and give the correct order of operations.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A highly accurate feature should be kept even when it would not exist during real use. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Apply the connected idea: Scaling without leakage

MAIN LESSON · SECTION 8

Practise counts, grouping and the evaluation claim

> Use the full A–F assignment table beside W2.

Supported group-versus-frame W2

Use workbook W2 for the connected example “Splits and related examples”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Match the grouping level to the question

For new trips on a familiar route, holding out whole trips addresses within-trip similarity. For new routes, hold out entire routes so all their trips stay together.

For forecasting future events, a time-respecting split may be needed: **training** on later events to predict earlier ones would not represent that intended use. The evaluation question determines the separation.

There is no universal percentage that makes a split valid. Large **group**s can make row counts unequal.

Check that the sets can support the intended comparison, including enough relevant cases and labels, without moving individual related frames just to hit a percentage.

This twelve-row example is too small to justify a reliable performance claim.

Challenge unique-row-means-independent reasoning

- Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the one-plant final test

- How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent plant-split task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Optional scaling validation example

Keep the **training** scale and k=1 fixed. Validation rows are (11,120,clear), (19,190,blocked), (12,80,clear). Their scaled feature pairs are (0.1,0.2), (0.9,0.9), (0.2,-0.2).

Nearest IDs are A,C,B, so **prediction**s clear,blocked,blocked score 2/3. The **training**-majority blocked baseline gets 1/3. These three examples do not establish general superiority.

The last brightness value scales to -0.2 because 80 is below the **training** minimum 100. New values can fall outside 0 to 1; do not refit or silently clip them.

Including validation features in minima or **range**s leaks evaluation information even without labels. **Fitting** a separate scale on validation also changes coordinates and breaks the intended comparison.

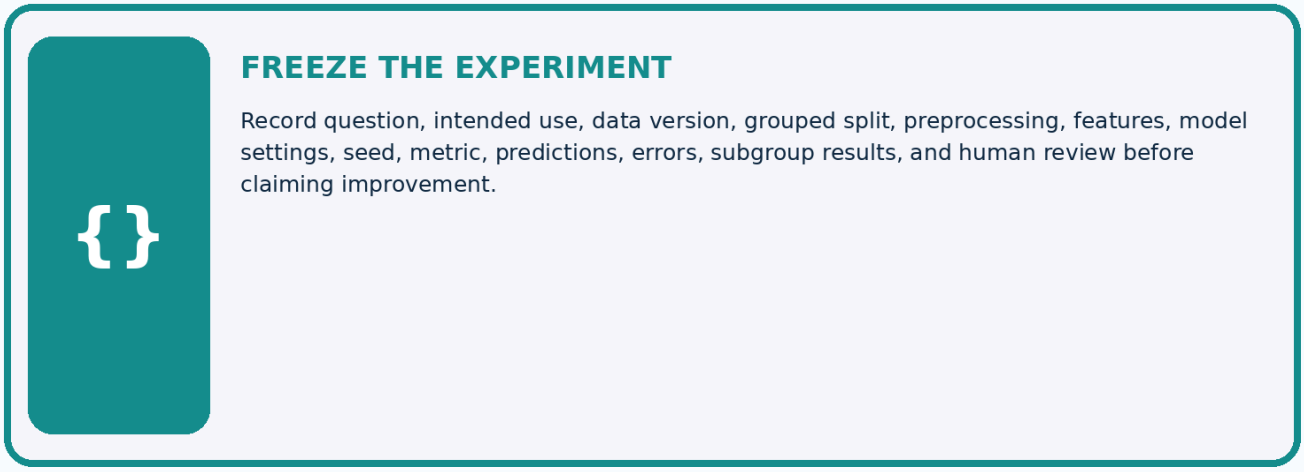
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before **group** discussion: Every **transform**ed value must lie between 0 and 1

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + **group**ed split + **leakage** check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + sub**group** slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: Scaling without leakage

Practise with less help: Distinguish frames from groups

Check your reasoning

Practise the connected topic: Scaling without leakage

Explain the picture

MAIN LESSON · SECTION 9

Investigate four supplied leakage paths and one allowed candidate

> Every promised detective field has an explicit prediction-time context.

Complete school-event timeline fields

Fictional school-event task: at the morning opening predict whether a queue exceeds ten minutes later	Availability / investigation
F1 · Number of desks open at prediction time	Available candidate input; still check collection and intended-use relevance.
F2 · Final queue category copied into filename after the event	Direct answer encoding unavailable at the prediction moment; exclude from prediction features.
F3 · Volunteer response logged after the long queue was observed	Later consequence; exclude from prediction-time inputs.
F4 · Same event photographed twice, one view in training and one in test	Related event crosses roles; group both views by event for a new-event evaluation.
F5 · Numeric scaling fitted on all event rows before splitting	Evaluation features influence fitted preprocessing; split/group first and fit only on training.

Normal, sparse-source, missing-ID and changed-goal probes

Complete split probe	Audit and bounded interpretation
Normal A–F assignment table	6/2/4 frames,3/1/2 groups,no shared listed trip; verify complete source membership too.
Boundary: final plant picture has only Fir	Three views but one new source plant; weak population evidence.
Missing metadata: G1 has no known trip ID	Do not invent an independent group; investigate source or document exclusion before evaluation.
Changed goal: evaluate new routes, while train/test share a route through different trips	Trip grouping alone does not match new-route claim; group at route level.

Audit, repair and record the learning boundary

1. Draw a prediction-time timeline for F1–F5, mark the answer-derived and later fields, then audit event identity and fitted scaling separately. For the full A–F trip plan, verify each source frame appears exactly once and group assignments do not cross roles. Recompute the supplied D2 move before repairing it by keeping all D frames together.

   Record a justified split before fitting; do not keep tuning a model on the final test. No real school data, model fit or score is supplied. Keep the selected five-plant split for the first individual attempt.

Keep evidence and observed status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Leakage Detective: draw a timeline for every suspicious field, inspect group identity, and repair the split before training.

1. Write the hypothesis and expected **evidence** before running.

2. **Leakage** Detective: draw a timeline for every suspicious field, inspect **group** identity, and repair the split before **training**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: repair evaluation across five plant identities

> Give a grouping rule that matches the unseen-plant claim.

Selected task and interpretation

Each of five plants has four photos. A random photo split puts different views of every plant in both **training** and **test set**s. Explain the problem and repair.

Assume the evaluation claim concerns photos of previously unseen plant identities, not additional views of already observed plants. The supplied task gives20 photos from5 plants but no exact frame assignments or model predictions. State a plant-level repair without inventing counts for a particular train/test allocation; keep all4views of each plant in one role.

Explain the small-source-count limitation and do not claim a measured accuracy change.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A feature records the expert's final species decision, made after the **prediction time**. Why is this **leakage**, and what should happen to it?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's information boundary reviewable

> Tie every leakage repair to a stated source relation or prediction-time rule.

Project artifact and checks

THIS WEEK'S CHECKPOINT Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.

Save the source version, prediction moment, allowed features, answer-derived/later fields, group definition and exact assignments. Record frame and group counts and any missing or excluded source records. Fit learned preprocessing only on training and preserve those parameters.

If a final test has guided development changes, label that exposure and obtain a fresh independent evaluation before making a final claim; a repaired split alone does not establish accuracy.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence, limits and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

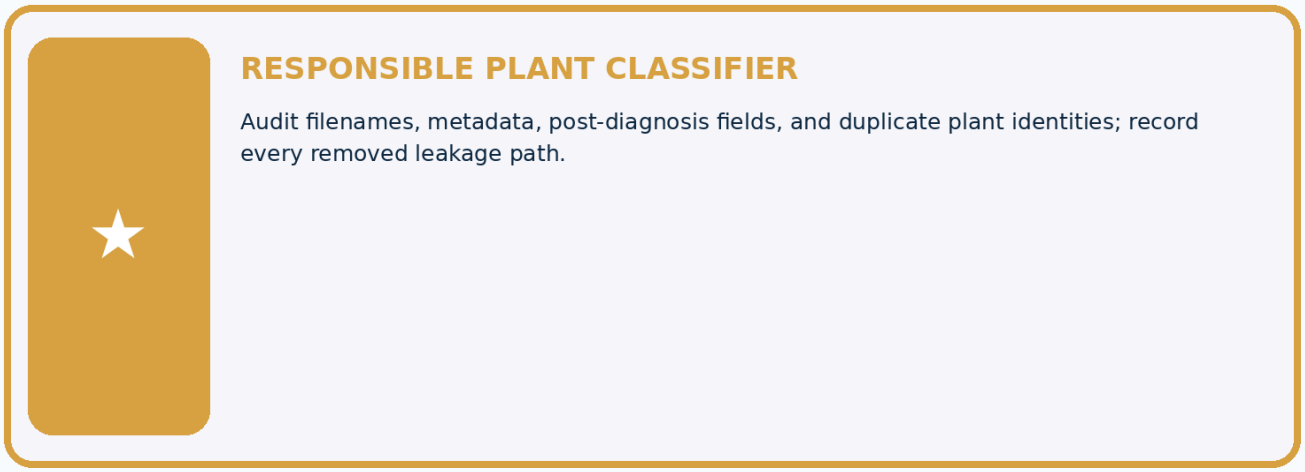
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish by naming the source or information boundary

> Explain what a corrected split can and cannot establish.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Choose a split unit that matches the intended **prediction** claim

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
leakage	target information improperly enters features	
prediction time	moment when real input is available	
post-outcome feature	field recorded after the target occurs	
split leakage	related records cross data splits	

Closer-explanation terms

Word	Meaning
Training set	Examples used to fit a model
Validation set	Separate examples used to compare choices during development
Test set	Held out examples used for final evaluation after choices are fixed

Word	Meaning
Group	Related examples that must stay together for a stated evaluation goal

▼ Extension vocabulary

Term	Meaning
Range	The largest training value minus the smallest for one feature
Fit	Learn and store values from training data
Transform	Apply a stored calculation to feature values
Leakage	Use information that should be unavailable while developing or evaluating a prediction method

▼ Earlier wording, alternatives and picture transcripts

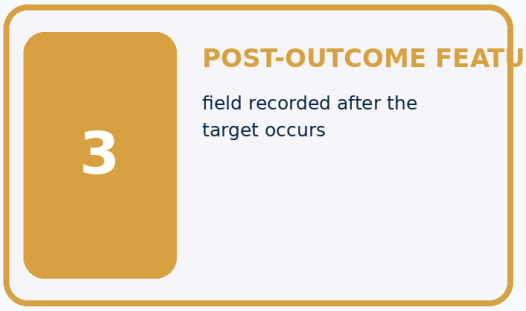
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A highly accurate feature should be kept even when it would not exist during real use.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Scaling without leakage

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **leakage**, **prediction time**, **post-outcome feature**, **split leakage**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A highly accurate feature should be kept even when it would not exist during real use.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

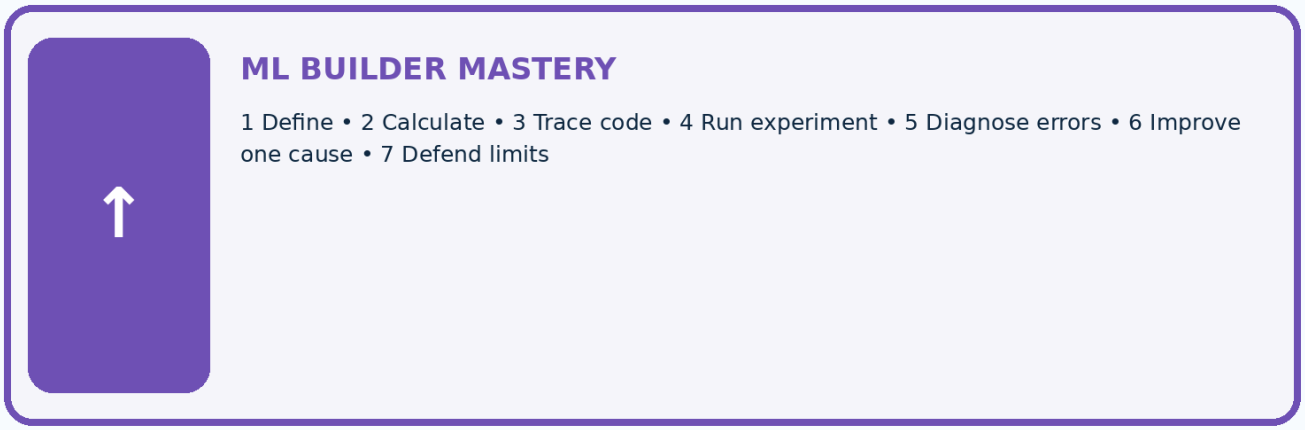
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected five-plant unseen-identity split task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

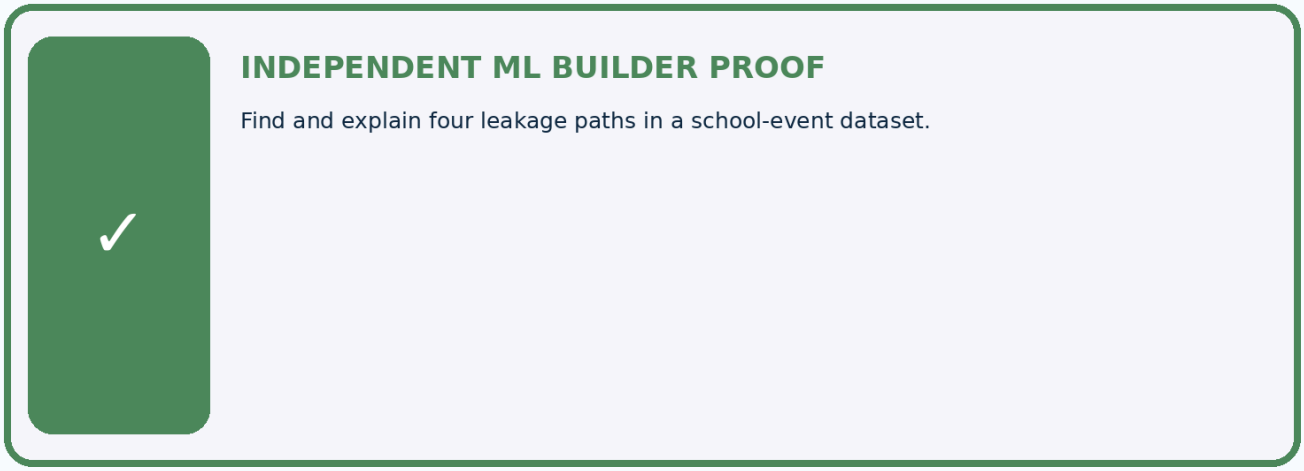
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Find and explain four **leakage** paths in a school-event dataset.

NEXT BRIDGE Carry the experiment **evidence** into: Dataset Cards.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Ten photos per leaf are randomly split so one leaf appears in train and test.

Explain and repair **leakage**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

Evidence or stage	Value or result
Leaf A, all views	10
Leaf B, all views	10
Leaf C, all views	10

▼ Earlier wording, alternatives and picture transcripts

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Keep related views in the same partition

Ten photos per leaf are randomly split so one leaf appears in train and test.

Explain and repair **leakage**.

Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Choose a split unit that matches the intended prediction claim	Explain the deciding step, show a trace or test, and state one limit.
Keep related examples together and audit group overlap	Explain the deciding step, show a trace or test, and state one limit.
Record split roles counts and limits before model evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Split by plant identity when testing generalisation to new plants.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Split by plant identity when testing generalisation to new plants.

- Which labelled step matches this detail? The final test probes one unseen plant, not three independent plants.
- 1. Assign **group**s
- 2. State claim
- 3. **Group** related views
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: State claim The final test probes one unseen plant, not three independent plants. Read the picture in order.

The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Move Fir's first view into **training** while retaining its other two in test. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The unseen-plant evaluation is compromised.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed explanation

After your attempt, compare with the supported explanation: The unseen-plant evaluation is compromised. Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Repair and keep the record

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
</details>

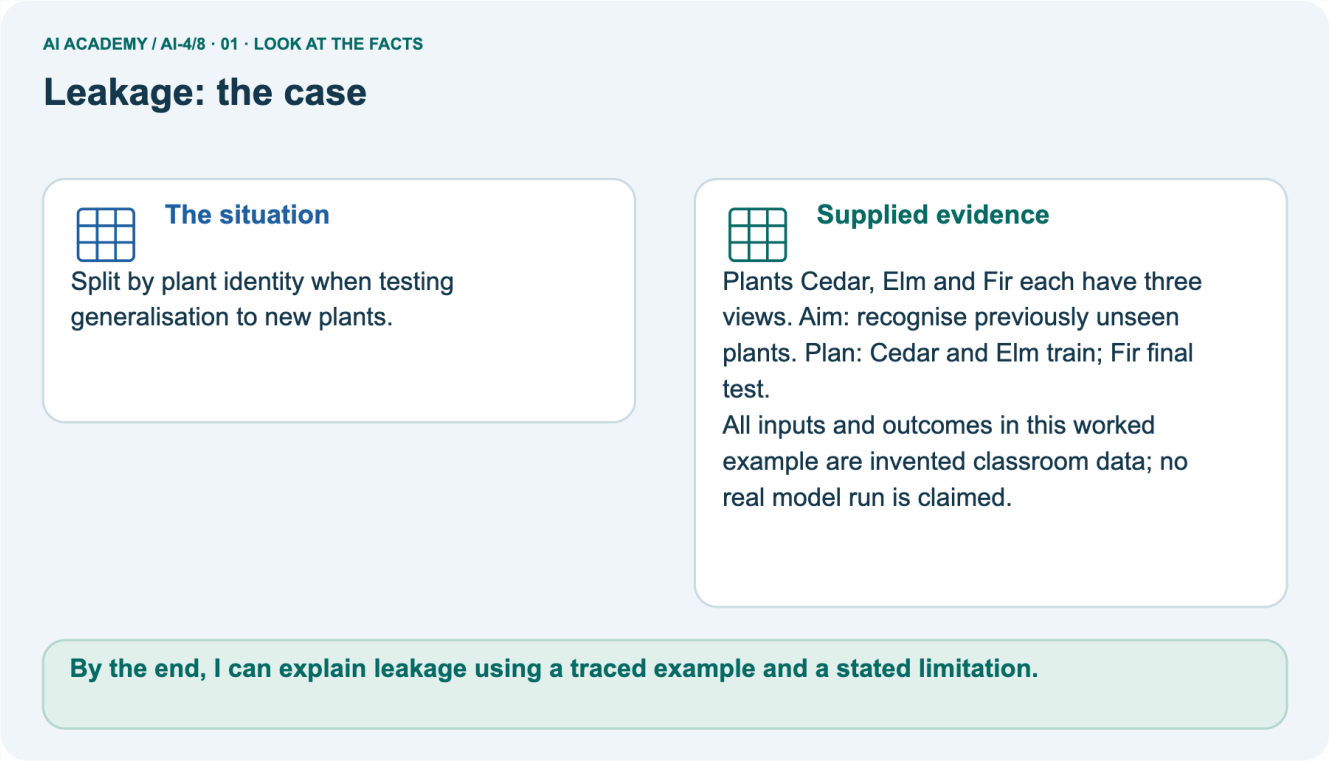
<details>
<summary>Reference · The six-picture journey</summary>
</details>

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Leakage: the case



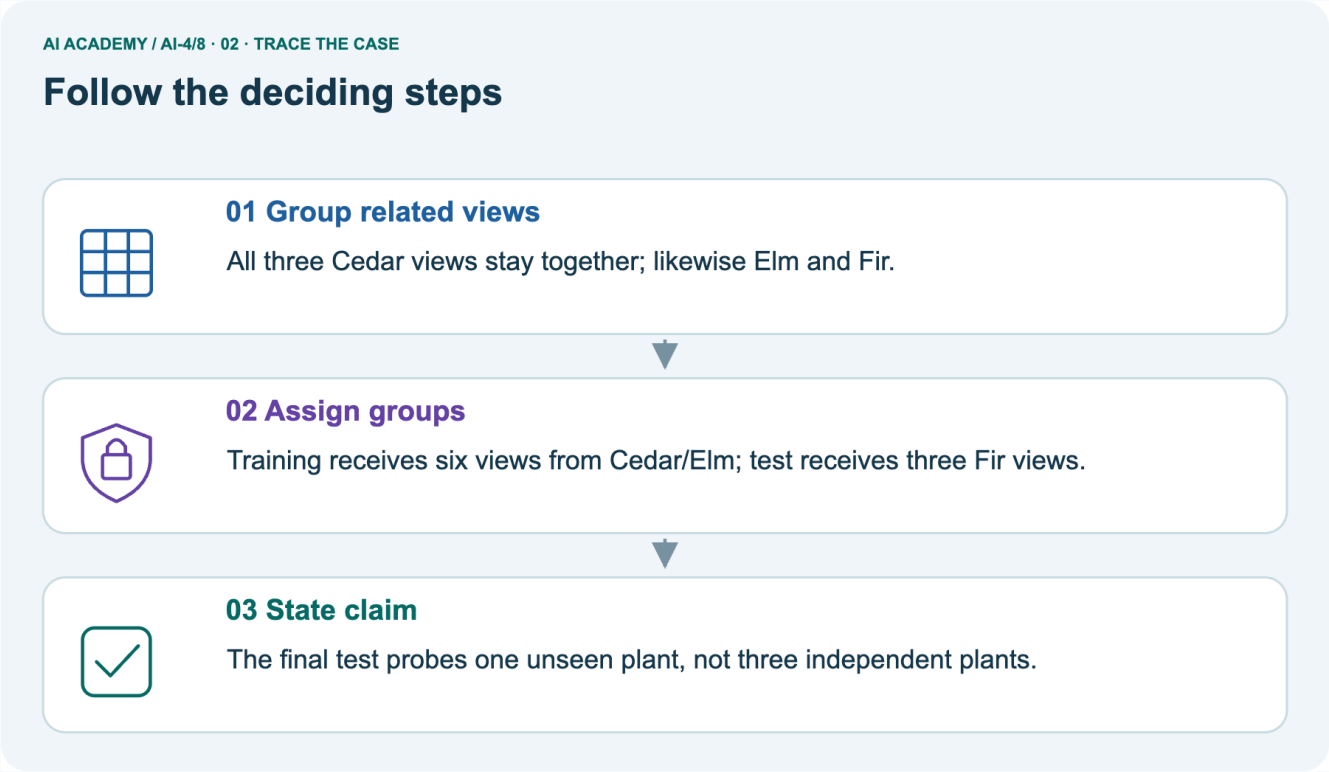
Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant identity when testing generalisation to new plants. Supplied facts: Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

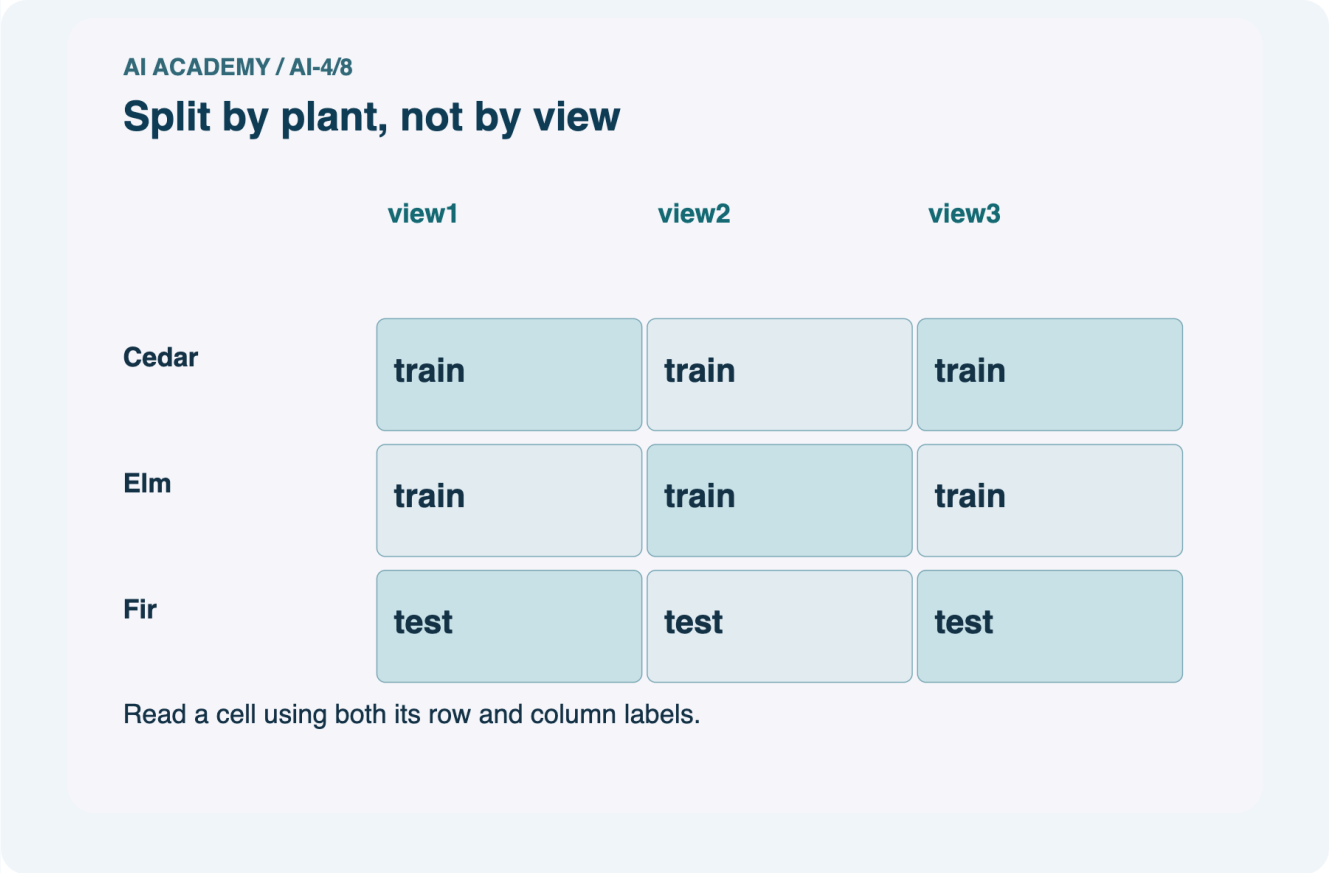


Follow the labelled picture with the matching explanation. Open the vector diagram

Group related views: All three Cedar views stay together; likewise Elm and Fir. → Assign **group**s: **Training** receives six views from Cedar/Elm; test receives three Fir views. → State claim: The final test probes one unseen plant, not three independent plants.

Point to each numbered stage and explain which recorded **input** it uses.

03. Split by plant, not by view

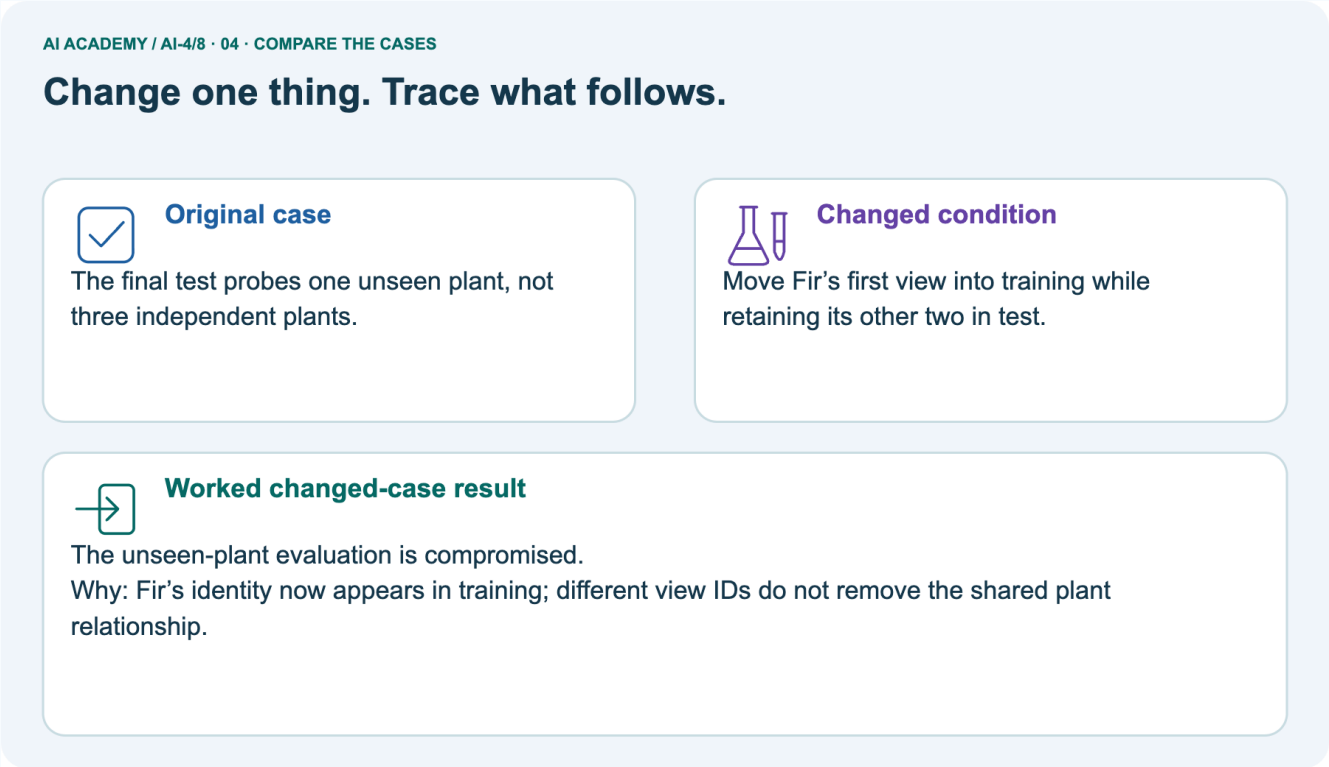


Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant, not by view. Cedar: view1 = train, view2 = train, view3 = train. Elm: view1 = train, view2 = train, view3 = train. Fir: view1 = test, view2 = test, view3 = test.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



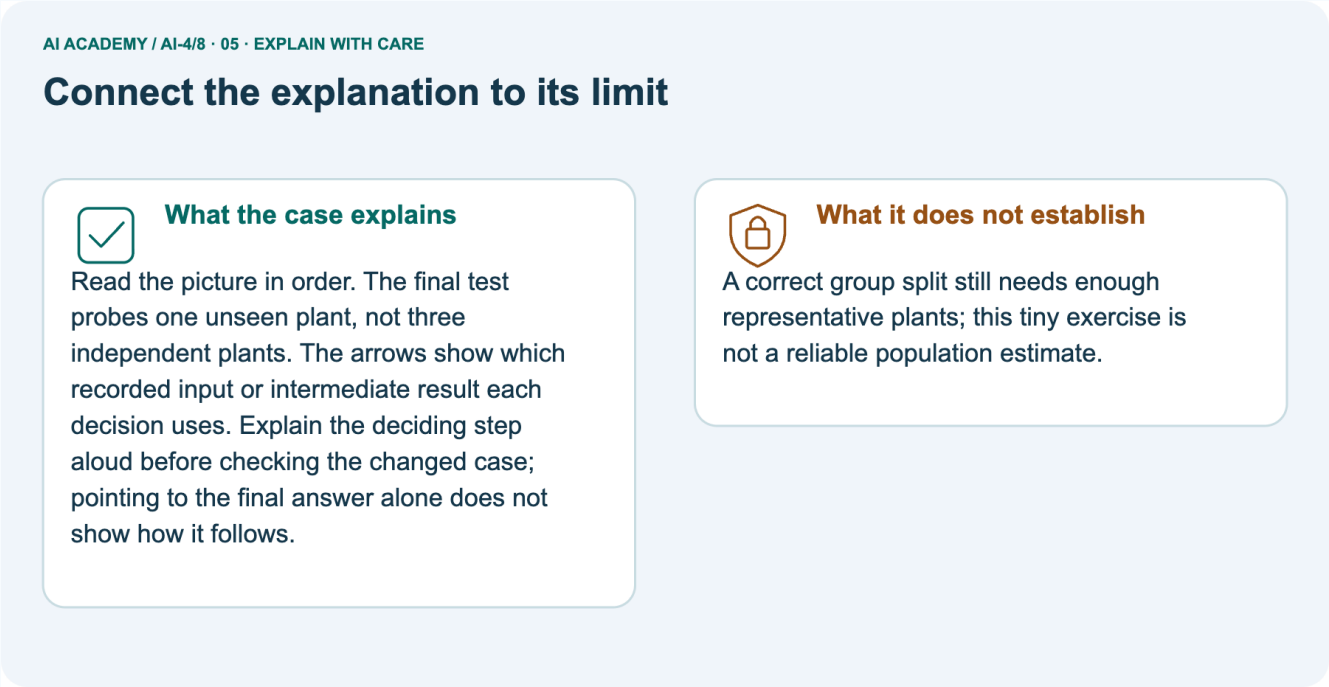
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final test probes one unseen plant, not three independent plants. Changed condition: Move Fir's first view into **training** while retaining its other two in test.

Expected result: The unseen-plant evaluation is compromised. Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

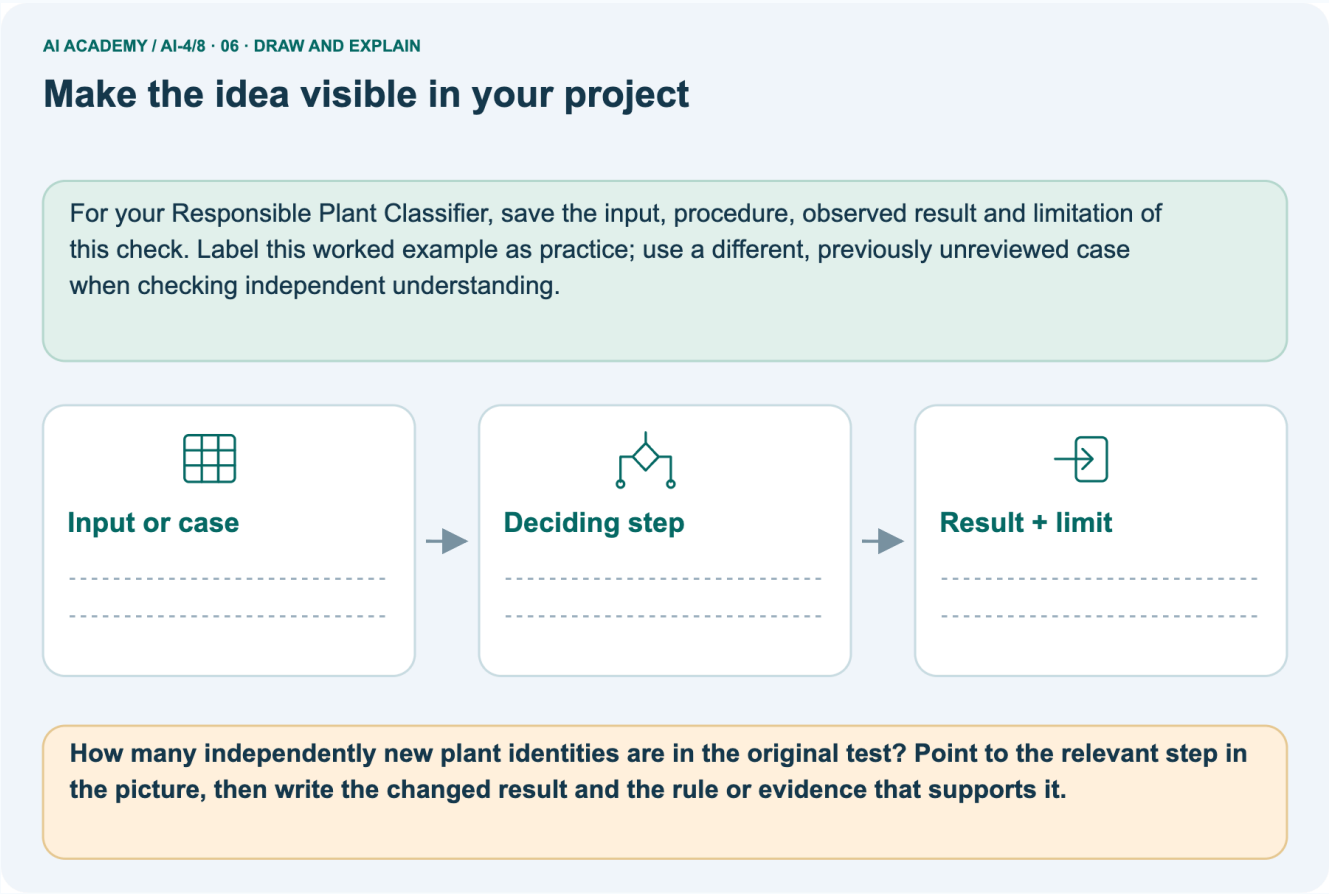
Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Leakage: the case

02. Follow the deciding steps

03. Split by plant, not by view

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.